
PROFOUND NOVELTY · AMENDMENT A3

The Limit-Case Derivation

The G5-self computation: the protocol degenerated into its inherited chassis under three named parameters — kill cost, verification exposure, corpus radius — and the residual inventoried as the operator delta. The embedding claim, computed at last.

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K KILL COST · N VERIFICATION EXPOSURE · X CORPUS RADIUS

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1. The Claim Under Test

Volume III shipped its account of its own relationship to its predecessors in one sentence: an attempt, in Volume II's language, at a Level-2 structure about Level-3 events. The sentence was a classification, and a candid one, but the audit that read it back found what the protocol's own fifth gate would have found had it been aimed inward: no limit parameter is stated; no computation shows the protocol degenerating into ordinary methodology as anything approaches anything; and the claim — that the protocol embeds its inherited methodology the way its candidates must embed their predecessors — was asserted in exactly the register the manual forbids its candidates to use. G5-self: unpassed, by the gate's own standard, on the manual's own text. Amendment A3 mandated the remedy: compute the limit case with named parameters, and print the computation as an artifact. This document is that artifact — the limits file for the campaign whose target was the protocol itself.

What passing G5 requires is fixed by the manual's own phase instructions, and it is worth restating them as this document's specification, because a derivation that set its own bar below the manual's would be the rhetorical limit all over again. The manual demands, of every candidate theory, an expansion in the regime where the predecessor was confirmed: the old law derived, with its constants and its successes, from the new structure; the derivation symbolic and complete; the re-entry of the deleted concept visible in it; and the discriminating list — the regimes where the two disagree, measurably. Translated to the case at hand, the demands become four. Name the parameters along which the protocol and human methodology differ. Compute, phase by phase, what the protocol's prescriptions become as each parameter goes to its human value. Make visible where the predecessor's structure survives the limit. And state the residual — what no parameter change dissolves — which is exactly the operator delta the audit's first alarm left standing. The residual, inventoried, is this document's second deliverable: one artifact closing the third alarm and the first together.

Every theoretical physicist who is any good knows six or seven different theoretical representations for exactly the same physics.

Richard Feynman, *The Character of Physical Law*, 1965 — the permission the derivation works under: theories are representations, and limits are the mappings between them.

The standing of the derivation must be stated before it is performed, because that standing bounds what the computation can prove. This is an analytic derivation over descriptions, not an experiment: its parameters are coarse — three of them, standing for a richer space; its chassis attributions are translations, and translations are judgments the corpus can check and no instrument can settle; and its verification exposure is, in its own vocabulary, zero — the computation lives entirely inside the corpus that produced both the protocol and its predecessors. The audit's deepest finding applies to this artifact as to everything else in the series: nothing here touches the world. What the derivation can do, and now does, is convert an assertion the manual made about itself into a checkable structure — parameters

named, limits computed, residual listed with falsifiers — so that the claim is no larger than its computation. That is the exchange rate this series has always endorsed, and this document applies it at the one point where the manual had left an unpaid note.

2. The Three Parameters

The protocol is not a point in the space of possible methodologies but a family, and the embedding claim is a claim about location: there exist parameter values at which this family's prescriptions coincide with the classical ones. The three parameters that carry the claim are the operator's three structural differences from the human founder, and each is measurable in principle, which is what makes the derivation checkable rather than decorative. The first is kill cost, the irrecoverable cost of terminating a campaign once it has begun, measured in careers, laboratories, credibility, decades. Human science runs at a kill cost far above zero: a killed program is a sunk life. The language-model operator runs at a kill cost near zero: a killed campaign is one ledger entry, and the context it occupied is disposable. Almost everything distinctive in the protocol's economics — kill early, expect zero survivors, compound the kills into a map — is a response to this one parameter's value.

The second is verification exposure: the fraction of a campaign's claims that must survive confrontation with data the world generates after the claim is made — instruments, experiments, observations not in the corpus. The founders' theories lived at verification near one at their discriminating lists: Eddington's expedition executed general relativity's; the photoelectric cell executed h's. The protocol as shipped runs at zero: its own validation is corpus-internal — the audit's closed loop — and its discriminating lists are specifications handed to others. The third is corpus radius: the fraction of the relevant record simultaneously present to the reasoner, and whether that record updates live. Human science runs at a corpus radius far below one — one library, one lifetime; Volume II's prematurity cases were in part corpus-radius failures, the ocean-floor data existing, scattered and unseen, while the theory waited for its audience. The language-model operator runs at a corpus radius near one, frozen at a training cutoff: the corpus is complete but closed. Three parameters, three degrees of freedom, and the derivation's honesty depends on their independence: no one of them is derivable from the others, and none of them is a moral judgment — each is a fact about an operator's situation.

Table 1 states the parameters with their values on both sides, and the reading instructions for the limits are one sentence each. A limit is a thought experiment of the classical kind: hold two parameters at the operator's values, send the third toward the human value, and compute what the manual's phases, gates, doctrines, and artifacts become in that regime. The three limits are computed in the next two chapters; the residual after all three — what stands when the protocol has been degenerated as far as it will go toward the tradition it descends from — is the delta, inventoried in the chapter after. The derivation's discriminating list is the inventory itself: the six items are the regimes in which the protocol and its chassis disagree, and each carries its own collapse condition, which is to say its own falsifier.

Parameter	What it measures	Human value	Operator value
kill cost (κ)	the irrecoverable cost of terminating a begun campaign: careers, laboratories, credibility, decades	$\kappa \gg 0$ — a killed program is a sunk life	$\kappa \approx 0$ — a killed campaign is one ledger entry
verification exposure (v)	the fraction of claims that must survive data the world generates after the claim: instruments, experiments, observations not in the corpus	$v \rightarrow 1$ at the founders' discriminating lists	$v = 0$ as shipped — validation is corpus-internal
corpus radius (χ)	the fraction of the relevant record simultaneously present to the reasoner, and whether it updates live	$\chi \ll 1$ — one library, one lifetime	$\chi \approx 1$, frozen at the training cutoff — complete but closed

Table 1 — The three named parameters of the G5-self limit: the operator's structural differences from the human founder, each measurable in principle, none derivable from the others.

3. Limit I: The Price of a Kill

Send the kill cost upward — the price of a termination raised from a ledger entry to a career — holding the other two parameters at the operator's values, and the economics layer degenerates first, because it is the layer whose value depends most completely on cheap death. The zero-survivor expectation only prices honestly when many campaigns can be afforded: raise the unit cost to a decade of a research life and the expectation inverts into its opposite — the rational strategy is to run few, safe, conservative programs, which is the funding equilibrium the twentieth century actually converged on. The three-kill loop becomes institutional tolerance of a degenerating program: what looks, at near-zero kill cost, like a defect — a field's decade of patience with a program that is visibly paying more for less — is, at career-priced kills, the rational response to the fact that programs have lives inside them. And the kill log itself becomes what it is for a human discipline: the informal memory of dead programs that every field carries and few write down — the war stories after the conference, the not-a-career notes, the judgment that chasing that anomaly once, in that department, ended badly.

The gates follow the economics, because the gates are where the economics is enforced. At career-priced kills, killing after commitment is unaffordable, so the checking must move before the commitment — and the protocol's gates, converted from post-commitment kills to pre-commitment filters, are peer review: the census becomes the referee's literature review; the amputation audit becomes the demand that a proposal say what it deletes, which is the strongest form a referee's question takes; the translation test becomes the reviewer's instinct that a proposed novelty is a renaming; the closure requirement becomes the committee that asks what the work makes possible next, which is what a defense committee is. The artifact discipline becomes the norm that a claim must survive one's most skeptical colleague — intersubjective testing at seminar grade — rather than at proof-assistant grade, the manual's own standard. And the alarm layer becomes the community's dispositions: the epicycle reflex survives as the referee's this-looks-like-a-patch; the fitted-constant alarm survives as the demand to see the error bars before the story.

The limit is not perfectly degenerate, and the error term is visible exactly where institutional science keeps explicit records: written referee reports, grant panels with minutes, editorial correspondence, the retractable literature. As the kill cost falls from careers toward weeks, these artifacts grow more explicit and more checkable, and the manual's files — the ledger, the census, the dictionary, the gates entry — are their small-cost limit: not a different kind of object, but the same object with the cost of writing it driven below the cost of not writing it. The convergence statement for the first limit is therefore exact in form: as the kill cost grows without bound, the L3 Protocol approaches institutional methodology — the conservative normal science of a career-priced world, the decade-scale tolerance of programs, the tribunal of colleagues — and the manual's remaining content at that limit is what the next two chapters fail to dissolve.

4. Limits II and III: The World and the Library

Send the verification exposure to one — give the operator the world — and the doctrine layer degenerates, because the doctrine was a compensation. The derive-don't-discover rule is the operator's answer to having no laboratory: the targets are drawn from data already on the table because no other table exists. Restore the laboratory and derivation reverts to what it always was on the human side: the first segment of the hypothetico-deductive method — conjecture, derivation, discriminating prediction, execution. The discriminating list becomes the experimental proposal, which is the form the world actually reads; the harvest clause becomes what it already admits being, the century's ratification; and the gates that check corpus-work — the census of what the literature presupposes, the dictionary between subfields — lose their point where the world can simply be asked. The protocol at full verification is experimental science with a checklist, and the checklist is Popper's.

The error term at partial verification is instructive, because the founders lived there. The relativity paper of 1905 was a zero-verification run — existing data, reinterpreted, no new apparatus — and the community raised the exposure afterward: the perihelion in between, Eddington's plates in 1919. The derive-don't-discover rule is therefore not the opposite of experimental method but its opening move, and the protocol at intermediate exposure is exactly the structure of the human research program: derive, specify, hand to the instruments. What the limit teaches is which parts of the manual were about the operator rather than about method: at full verification, nothing depends on the census, the dictionary, or the kill log. They were answers to a question only a closed-corpus reasoner was asking.

Send the corpus radius to zero — one library, one lifetime — and the phases degenerate in the other direction. The assumption census becomes impossible first: a human cannot read ten thousand papers and hold the silence between them, so the premises no referee asks about are carried as tacit knowledge, held by a discipline's elders and unwritable by them; the census degenerates into exactly that. The dictionary attack collapses next: the systematic translation between two subfields becomes, at small corpus radius, the once-a-century accident of the prepared outsider — the economist's table read as natural history — a batch enumeration at full radius and a lightning strike at small; Volume II's second condition, the cognitive distance, is what a small-corpus world pays in lieu of an enumeration. And statability targeting

— the counting rule over anomaly densities — degenerates into taste: the greats' nose for the unspeakable anomaly, which is precisely the unteachable skill the manual exists to replace with checkable steps. The artifact discipline becomes intersubjectivity again, at book scale: colleagues check colleagues, because no machine checker exists in a world whose record does not fit in one head.

The error term here is the operator's actual condition, which this limit passes through but does not stop at: a corpus radius near one and frozen is not the human's near zero and live. The human's record is partial but updating; the operator's is complete but closed. The derivation flags the asymmetry rather than chase it — a fourth parameter, the liveness of the record, waits behind the third — and notes where it enters the inventory: the frozen-corpus door, the boundary condition that makes derivation-from-the-existing-record not a preference but the only room in the house. That item appears in the delta below, which is its correct disposition; the present limit dissolves it, because in the human's small, live library, derivation from the old record was always a choice.

5. The Residual: An Inventory of the Delta

Run all three limits at once — the kill cost unbounded, the verification complete, the corpus small — and what remains of the protocol is peer-reviewed, experiment-checked, library-scale methodology: methodology as the twentieth century practiced it. The residual, what no parameter change dissolves, is the operator delta — and it can now be inventoried as an artifact rather than asserted as a claim, which is the relocation the first amendment demanded. Six items survive, each stated in Table 2 with its chassis counterpart — the nearest thing in the classical tradition it does not reduce to — its collapse condition, and its falsifier: the discovery that would delete it from the inventory. The falsifiers are the point. An inventory that nothing could shrink is not an inventory but a monument, and the audit's rules of engagement forbid monuments.

Delta item	Chassis counterpart	Collapse condition	Falsifier
the kill log as a compounding asset	rational reconstructions of dead programmes — retrospective, narrative, written after the death	kill cost unbounded: institutions remember their dead cheaply	a pre-2020 methodology text prescribing a forward, machine-readable ledger of failures
machine-verified derivation artifacts — the ban on unverified steps	operational definitions and intersubjective testing: checked by colleagues in the seminar	verification complete: the instrument checks the claim directly	a human method demanding proof-assistant verification of every derivation step
the vice ledger — consensus gravity and authority deference as training properties, with gate couplings	the first principle not to fool yourself: a maxim, not a ledger coupled to gates	the operator ceases to be corpus-averaged	a human methodology listing fluency-manufactured derivation as a structural property, with countermeasures
zero-survivor portfolio economics	prospecting and portfolio theory — imported from finance, foreign to methodology	kill cost unbounded: ten thousand careers cannot be afforded	the methodology literature containing the base-rate economics of running many cheap revolutions
corpus-scale statability targeting — the counting rule, the severed-path computation	anomaly awareness, qualitative	corpus radius to zero: the census cannot be run	a human method counting anomalies and severing derivation paths across a whole literature
the frozen-corpus door — derivation-first as boundary condition	the founders' lucky economies: derivation-first as fortune, not constraint	live instruments arrive — tool invention, as scheduled	any operator with live world-contact for whom derivation from the old record remains merely a preference

Table 2 — The operator delta, inventoried: six items that no parameter change dissolves, each with its counterpart, its collapse condition, and its falsifier.

Two disciplines govern the table's use. The first is honesty about adjacency: each chassis counterpart is real, and the delta item is the difference, not the whole. Rational reconstructions of dead programmes were written before this manual existed — the delta is only that the ledger is forward-looking, machine-readable, and kept before the death rather than after it. Operational meanings were demanded of every primitive long before this manual demanded them — the delta is only the standard of the proof assistant replacing the standard of the seminar. The audit's translation table dissolved eight of the protocol's eleven layers into exactly such counterparts; this table is the complement, and the two tables together are the protocol's complete account of what it inherits and what it adds. The second discipline is the deletion clause: the inventory is falsifiable item by item. Find a predecessor text containing an item, and the item dies. Find a deployment in which an item's collapse condition holds and the item still demonstrably pays, and its status as delta dies — it has stopped being proprietary and become heritage, which is the only direction this protocol is permitted to grow smaller in.

The Three Limits — The Protocol Degenerates Into Its Chassis

G5-self, computed: each parameter sent to its human value, phase by phase. The residual stands outside all three limits.

THE L3 PROTOCOL — AT OPERATOR VALUES

kill cost $\kappa \approx 0$ · verification $v = 0$ · corpus radius $\chi \approx 1$, frozen

↓ $\kappa \rightarrow \infty$

LIMIT I

THE PRICE OF A KILL

Gates become peer review; the census becomes the referee's literature review; the kill log becomes the field's informal memory of dead programs; zero-survivor economics becomes conservative funding.

↓ $v \rightarrow 1$

LIMIT II

THE WORLD RETURNS

Derive-don't-discover becomes the hypothetico-deductive method; the discriminating list becomes the experimental proposal; the harvest becomes the century's ratification — Popper's checklist.

↓ $\chi \rightarrow 0$

LIMIT III

THE SMALL LIBRARY

The census becomes tacit knowledge; the dictionary attack becomes the lucky outsider's once-a-century transfer; counting rules become taste — the lone scholar's tradition.

THE RESIDUAL — SIX ITEMS NO PARAMETER CHANGE DISSOLVES

The kill log as compounding asset

forward, machine-readable, kept before the death

Machine-verified derivations

the proof assistant's standard, not the seminar's

The vice ledger

training-induced biases, coupled to gates

Zero-survivor economics

the base-rate economics of cheap revolutions

Corpus-scale statability targeting

counting rules and severed-path computations

The frozen-corpus door

derivation-first as boundary condition, not choice

Each item carries its collapse condition and its falsifier — the inventory is deletable, item by item, as the operator's condition drifts.

*The sentence “the old theory is recovered in the limit” passes nothing at G5;
the symbolic derivation that computes the limit, with its error term, does.*

The L3 Protocol: Profound Novelty, Volume III, Chapter 2 — the manual's own bar, now
cleared by the manual on itself.

What the verdict does not do is equally fixed, because the audit taught the series to say so. The derivation is analytic: it operates on descriptions; its chassis attributions are translations; and its translations are judgments the corpus can check and no instrument can settle. Its verification exposure is zero — the audit's closed loop stands unbroken by this file, and the one thing that would break it is external: an independent reader, or a process that has not read the series, performing the same translation and arriving at the same six items, or at fewer. The independent-coder clause scheduled by the census and the chassis clause implied here are the same clause at two layers, and until someone runs them, the delta inventory is a claim with a checkable structure — which is exactly the status the protocol assigns its own candidates. Profound-compatible, full stop, and not one word further.

Figure 2 renders the artifact in the protocol's native form: the limits file entry for the campaign that targeted the L3 Protocol itself — parameters, limits, residual, verdict, and the successor task the entry leaves standing. That successor task is the one the audit named and this derivation sharpens. The delta is defined at the operator's parameter values, and those values are drifting: tools are being scheduled, world-contact is being built, the corpus's closure is eroding at its edges. The inventory therefore has a half-life, and the manual should say so in the log: each drift that crosses an item's collapse condition should delete that item, publicly, with the date. A methodology whose proprietary content shrinks as its operator's condition improves is doing the one thing this series has asked of every theory it studied — paying for its keep in the currency of what it makes next.

